Linjun Yang

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Dr. Linjun Yang is a result-driven leader, who enjoys solving challenging problems, shipping products, and growing people. He has ∼20 years of experience across algorithms and engineering in building machine learning and search engine technologies. He is among the first shipping deep learning techniques and LLM-as-judge in large-scale search engine to improve search. He has extensive experience managing and growing machine learning science and engineering teams, applying AI to improve productivity, and delivering highest quality products and platforms through deep, principled, and scalable machine learning and search technologies.

PROFESSIONAL EXPERIENCE

Shopify Dec. 2024 - Oct. 2026

VP of Search and MLE Discipline Lead

Led Search and Catalog, with end-to-end responsibility for search metrics, relevance, infrastructure, catalog intelligence, and foundation models.

- **Modernized Search Platform.** Led the M&A process for a search team and integrated the team and its technology into Shopify within six months; subsequently increased platform throughput by 20×.
- **Search Relevance.** Built a modern relevance stack combining hybrid embedding and lexical retrieval with DNN reranking. Shipped generative retrieval and personalized query rewriting in production with latency under 30 ms.
- **ML Foundations.** Led development of Shopify's foundation models for embeddings, generative retrieval and recommendation, alongside Tangle and Tangent for ML development.
- **Shopify Catalog.** Built from scratch and then scaled Shopify Catalog to billions of products and integrated with partners including OpenAI and Meta. Built catalog intelligence spanning product clustering, taxonomy evolution and prediction, and a commerce knowledge graph.

Microsoft Bing Mar. 2020 - Dec. 2024

Vice President and Chief Scientist, Core Search and AI

- leading Core Search and AI engineering team (∼200 people world-wide), responsible for Web relevance (from metric to serving stack, from retrieval to ranking), caption / experience, personalization / recommendation, and whole page optimization.

Transform Search with LLM

- Be among the world-first by launching a GPT-based web search metric and distilled relevance stack from GPT, resulting in the largest relevance improvement in Bing in two decades (https://tinyurl.com/bingptlaunch).
- Modernized search relevance using LLMs, including LLM-based ranking and offline ranking, LLM-based query understanding, and LLM-based document understanding (synthetic clicks).
- Pioneered generative search and delivered multiple generative search products, such as deep search, generative captions/QnA, and generative search.

Relevance Fundamentals

- Enhanced search relevance stack through continuous deep learning innovation, including BERT-based ranking models, embedding-based retrieval with vectorization of web documents (FBV), DL-based caption and QnA models (DeepExtract, DeepCaption), leading to superior relevance from Bing in comparison with Google
- Ship term-sharded indexing in Bing (Fluid) with multiple rounds of retrieval improvements, which lays the solid foundation for search relevance.

- Developed BGM (Bing Generative Model) as the foundation for SLM distillation, widely adopted by multiple teams, including Ads. Successfully shipped the first SLM application in WebXT and drove multiple rounds of model iterations. Shipped multiple iterations of BingLR models as the foundation models for DL-based models in core search and adopted by other teams.
- Implemented full-body vectorization, an embedding model for encoding queries and web pages, which unified for all web search scenarios, and has been adopted by multiple teams to improve business metrics across multimedia, feeds, news, local, ads, office, and more.

Innovation Beyond Core Search

- Launched Griffin Go Big, leveraging web retrieval techniques to enhance Ads, resulting in $100M in ads revenue.

Others

- Build personalization and recommendation into Bing search from scratch.
- Drive fundamentals and improve team's engineering culture, establish the relevance iteration framework / process, modernize ML development based on AML, build engineering culture in the team. Grew multiple partner-level eng managers.

Facebook Oct. 2017 - Mar. 2020

Senior Engineering Manager

- Build search core machine learning team from scratch to 10+ people and lead the team working on cutting-edges deep learning techniques and platform.

- Groups search - 2019 H2
  Lead Groups search ranking team, improve Groups search quality.
- Embedding based retrieval - 2017 - 2019
  Initiate embedding-based retrieval project and deploy the technique to Facebook search to leapfrog semantic matching.
- Search ranking - 2017 - 2019
  Improve search ranking stack quality through techniques including training data mining, deep learning modeling techniques, stack optimization, feature engineering, etc.
- Search ML platform - 2019
  Build Machine Learning platforms to improve engineer productivity including training data pipeline and feature engineering platform.
- Search metric design - 2019 H1
  Define search quality metrics for Search relevance.

Microsoft Bing Jul. 2012 - Oct. 2017

Principal Software Engineer Manager

- build image understanding team from the scratch to 10+ people and lead the team to research and develop cutting-edge computer vision and machine learning technologies to improve various products.

- Deep learning for image understanding
  Research/develop deep neural network models for improving various image understanding tasks including feature learning, image categorization, semantic embedding, etc.
- Visual search (a.k.a search by image)
  Build visual search product e2e including defining measurement set/metric, developing models/rankers to improve the user experience of visual search, developing backend, and working with partners on UX / pipeline.
- Large-scale duplicate/similarity search system
  Build a system to retrieve similar/duplicate images from billion-scale database for billions of images per day.
- Image tagging
  Build algorithms/system to generate tags for any given image based on deep learning and similarity

search.

Microsoft Bing Nov. 2011 - Apr. 2012
Visiting Researcher, Multimedia Search Team

- Designing and implementing back-end image processing library/pipeline.
- Developing billion-scale image duplicate detection system.

Microsoft Research Asia Jul. 2006 - Jul. 2012
Associate Researcher, Media Computing / Internet Media Group

- Visual search
  We study various methods to improve the precision of image and video search by example.
- Content-based Image Ranking/Reranking
  This objective of this project is to improve image search ranking by utilizing visual content analysis.
- Video analysis SDK
  Develop a suite of libraries for image and video content analysis based on Windows DirectShow API.

EDUCATION

Ph.D, Multimedia Information Retrieval
Delft University of Technology, The Netherlands 2013

M.S., Compute Science
Fudan University, Shanghai, China 2006

B.S., Electronics & Information System
East China Normal University, Shanghai, China 2001

AWARDS

The Best Paper Award at ACM International Conference on Multimedia (ACM MM), Beijing, China, 2009.

The Best Student Paper Award at the 18th ACM Conference on Information and Knowledge Management (CIKM), Hong Kong, 2009.

PUBLICATIONS

https://scholar.google.com/citations?user=cvgKxDQAAAAJ